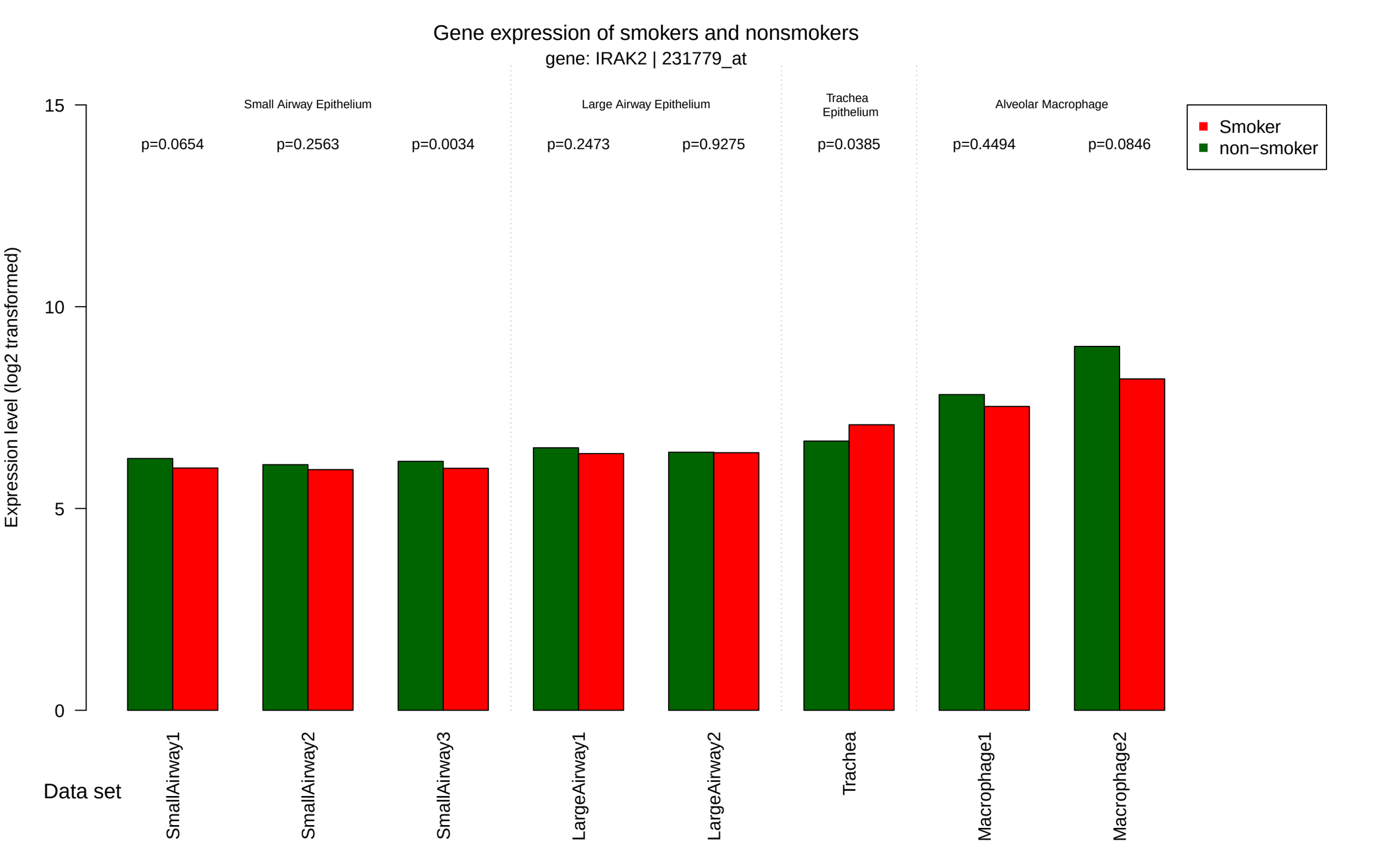
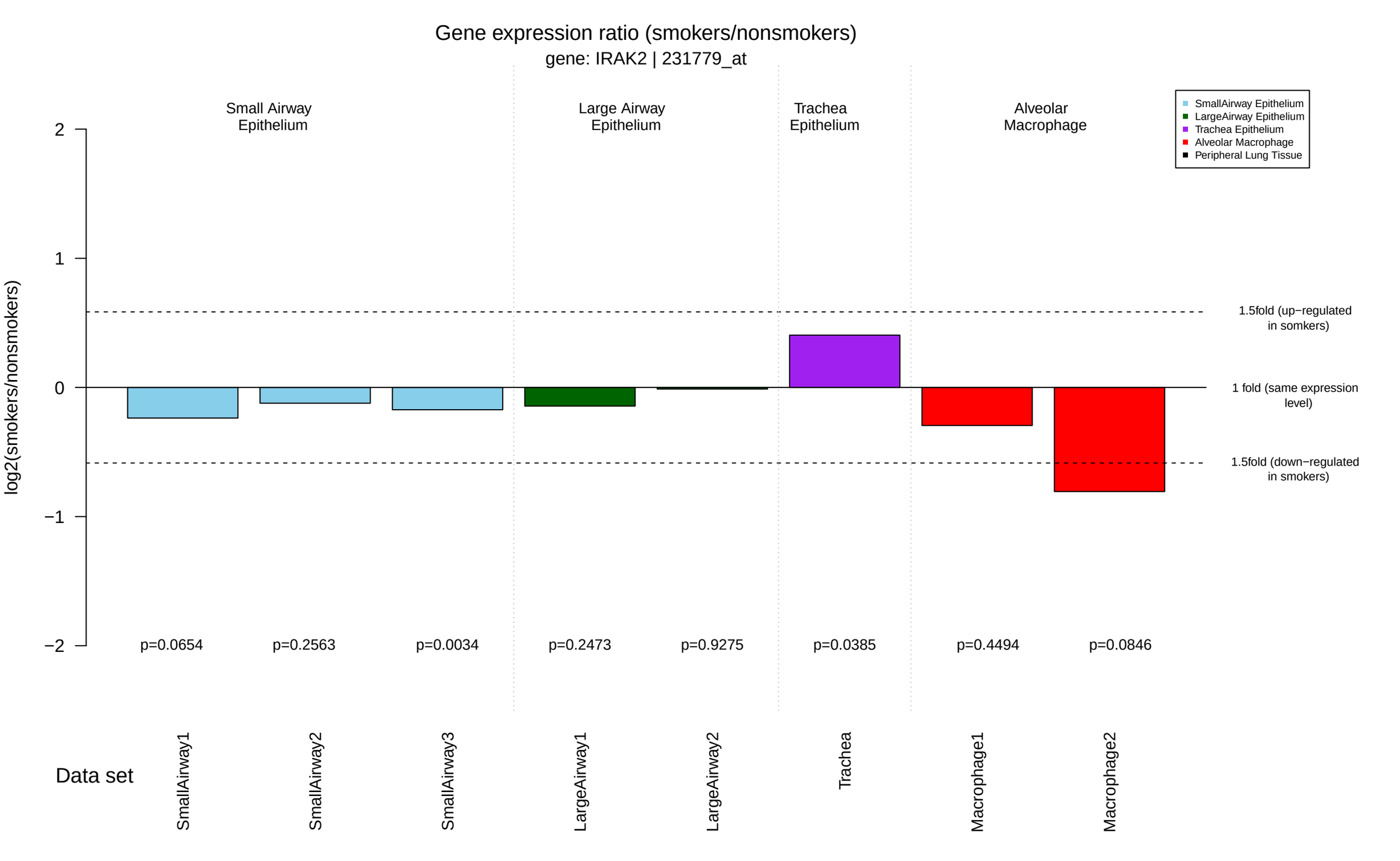
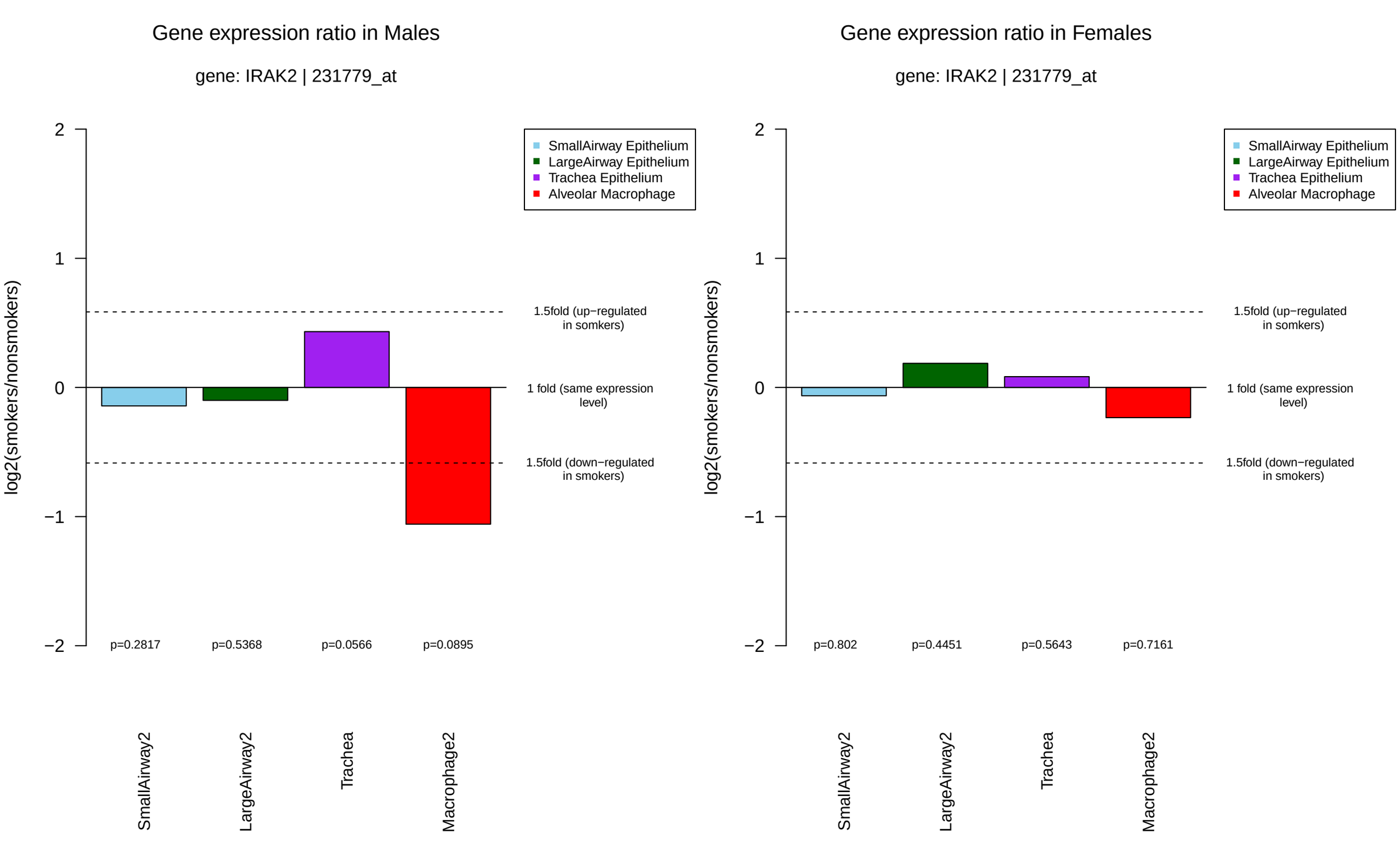
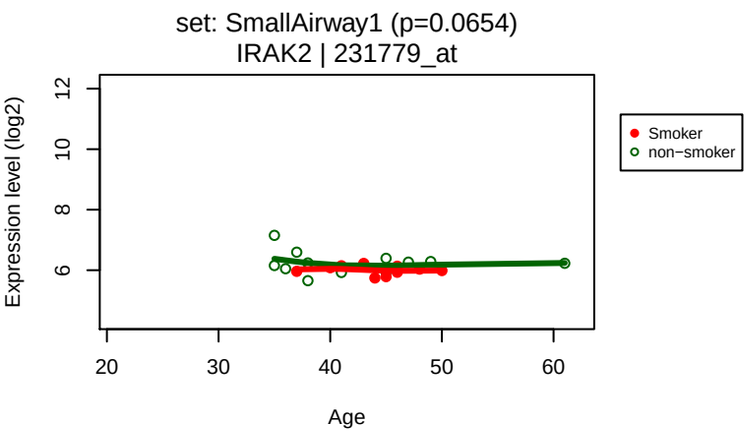
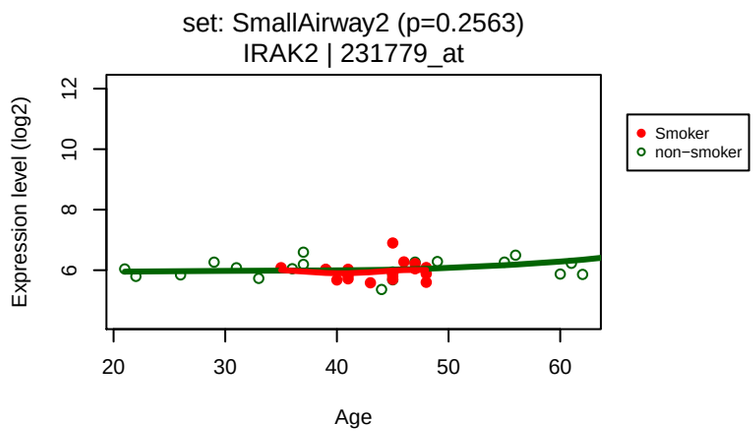
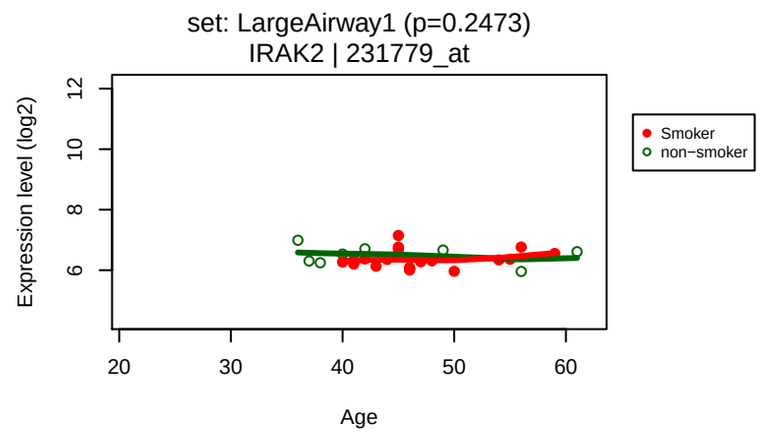

Gene expression of smokers and nonsmokers
gene: IRAK2 | 231779_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0654
p=0.2563
p=0.0034
p=0.2473
p=0.9275
p=0.0385
p=0.4494
p=0.0846
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

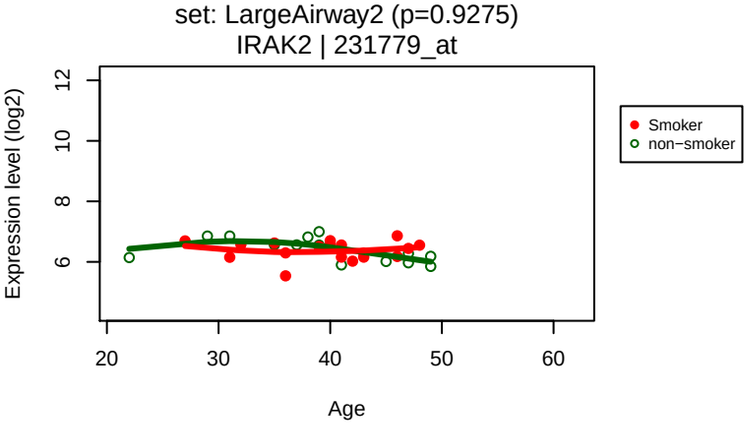

Gene expression ratio (smokers/nonsmokers)
gene: IRAK2 | 231779_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0654
p=0.2563
p=0.0034
p=0.2473
p=0.9275
p=0.0385
p=0.4494
p=0.0846
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

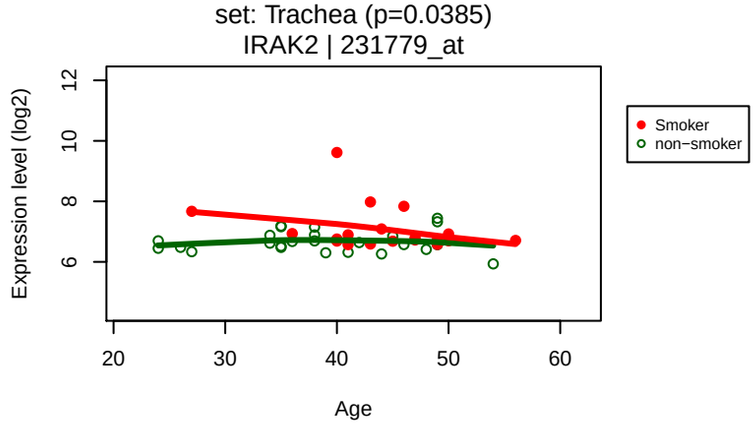

Gene expression ratio in Males
gene: IRAK2 | 231779_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2817
p=0.5368
p=0.0566
p=0.0895
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRAK2 | 231779_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.802
p=0.4451
p=0.5643
p=0.7161
SmallAirway2
LargeAirway2
Trachea
Macrophage2

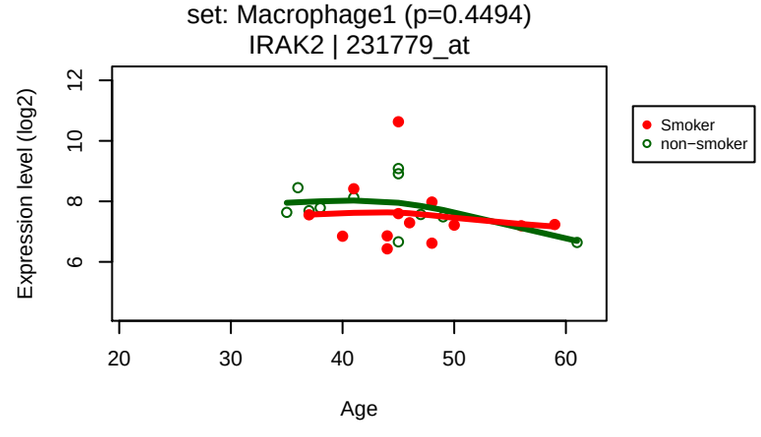

set: SmallAirway1 (p=0.0654)
IRAK2 | 231779_at
Expression level (log2)
Age
Smoker
non−smoker

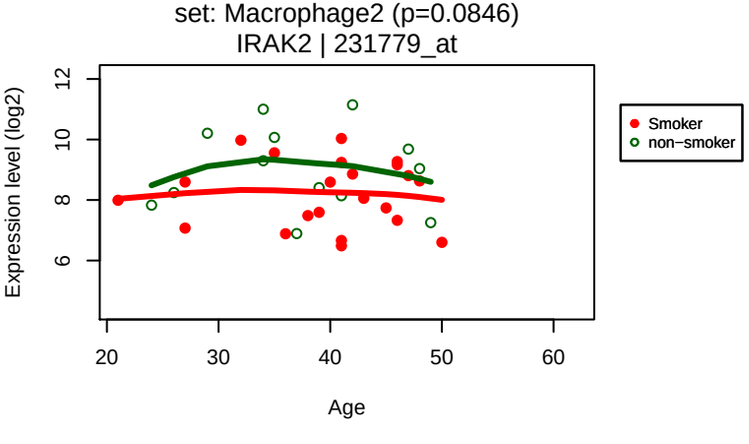

set: SmallAirway2 (p=0.2563)
IRAK2 | 231779_at
Expression level (log2)
Age
Smoker
non−smoker

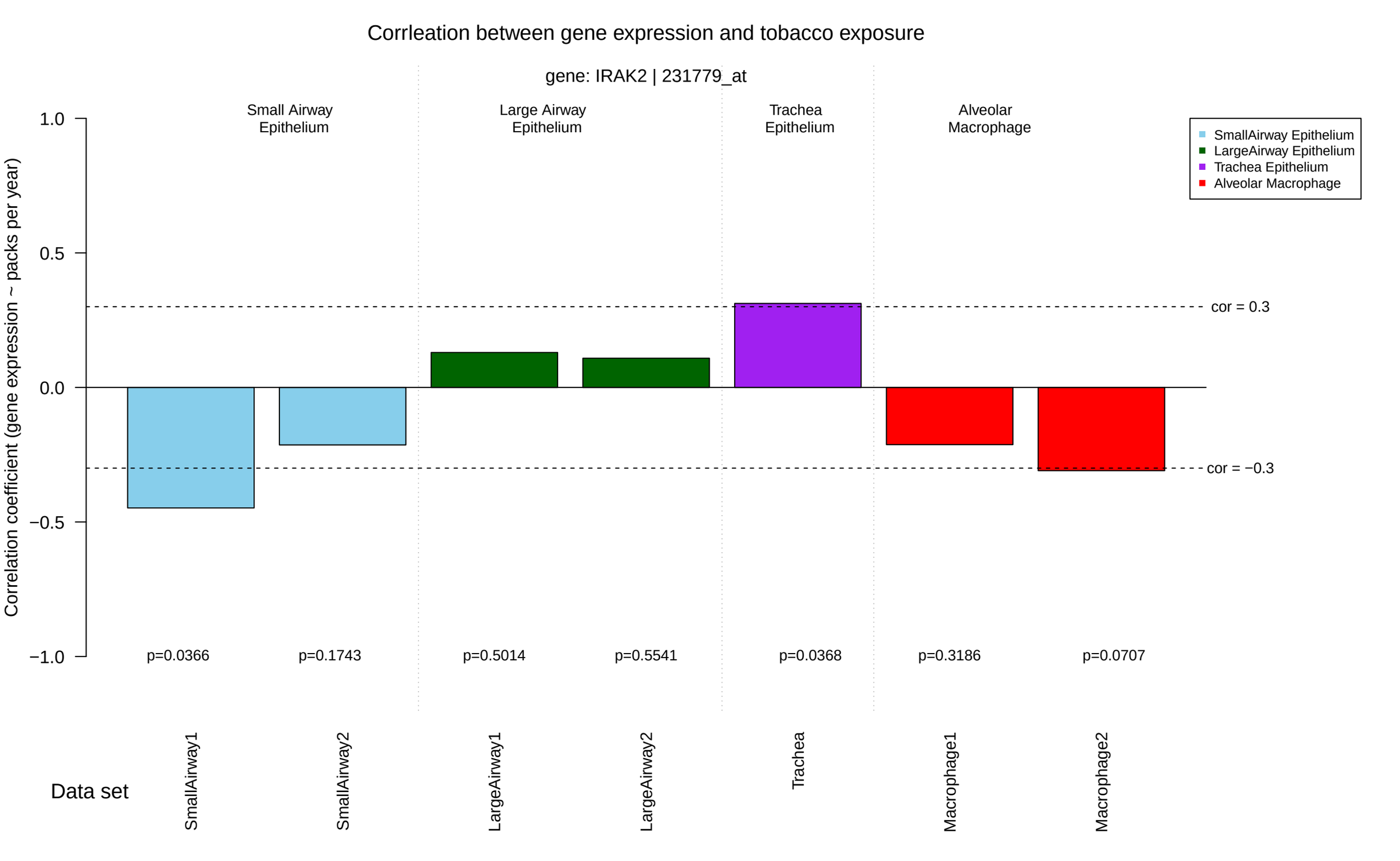

set: LargeAirway1 (p=0.2473)
IRAK2 | 231779_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9275)
IRAK2 | 231779_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0385)
IRAK2 | 231779_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4494)
IRAK2 | 231779_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0846)
IRAK2 | 231779_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRAK2 | 231779_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0366
p=0.1743
p=0.5014
p=0.5541
p=0.0368
p=0.3186
p=0.0707
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2